

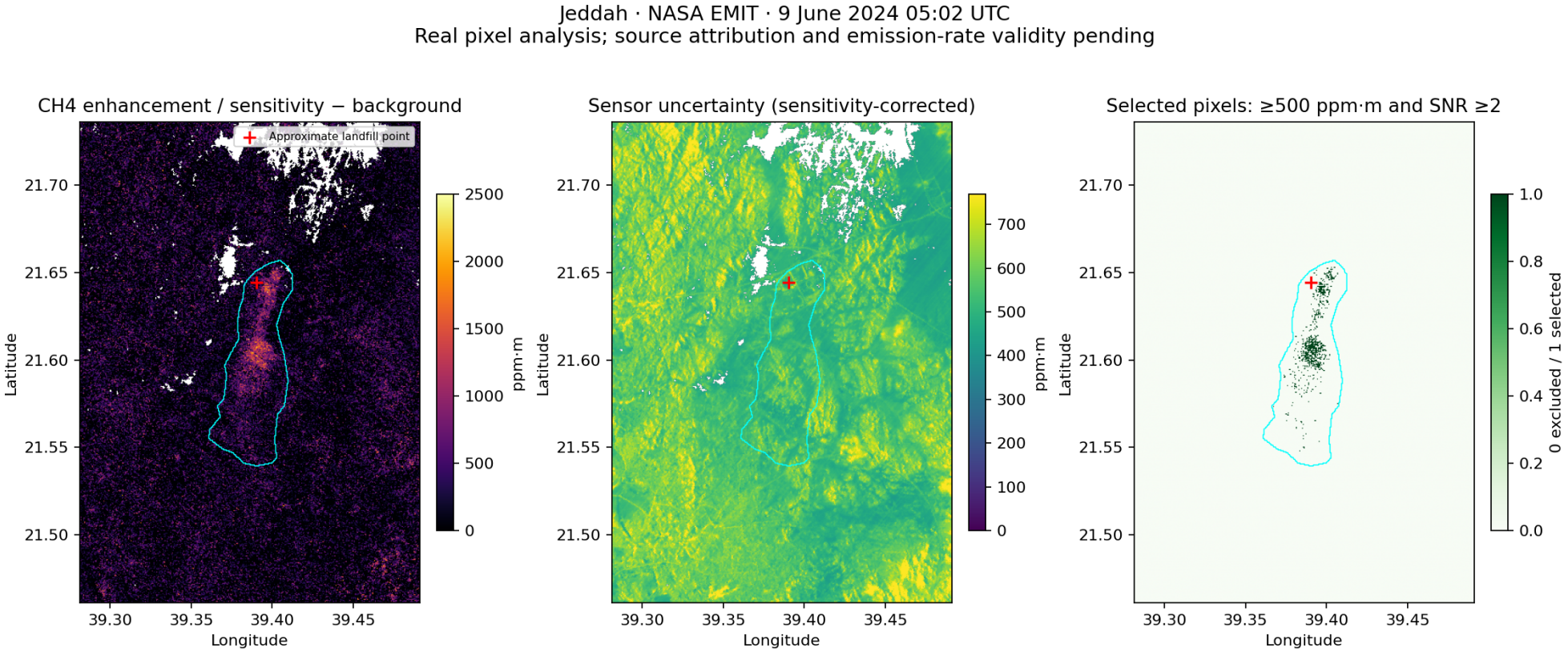
# EcoOrbit | methane and carbon intelligence

- Team: EcoOrbit | Challenge: Air Intelligence
- Real NASA methane pixels and separate published CO2 references
- Site ML and multispectral analysis support follow-up

## **Solution: three real methane observations**

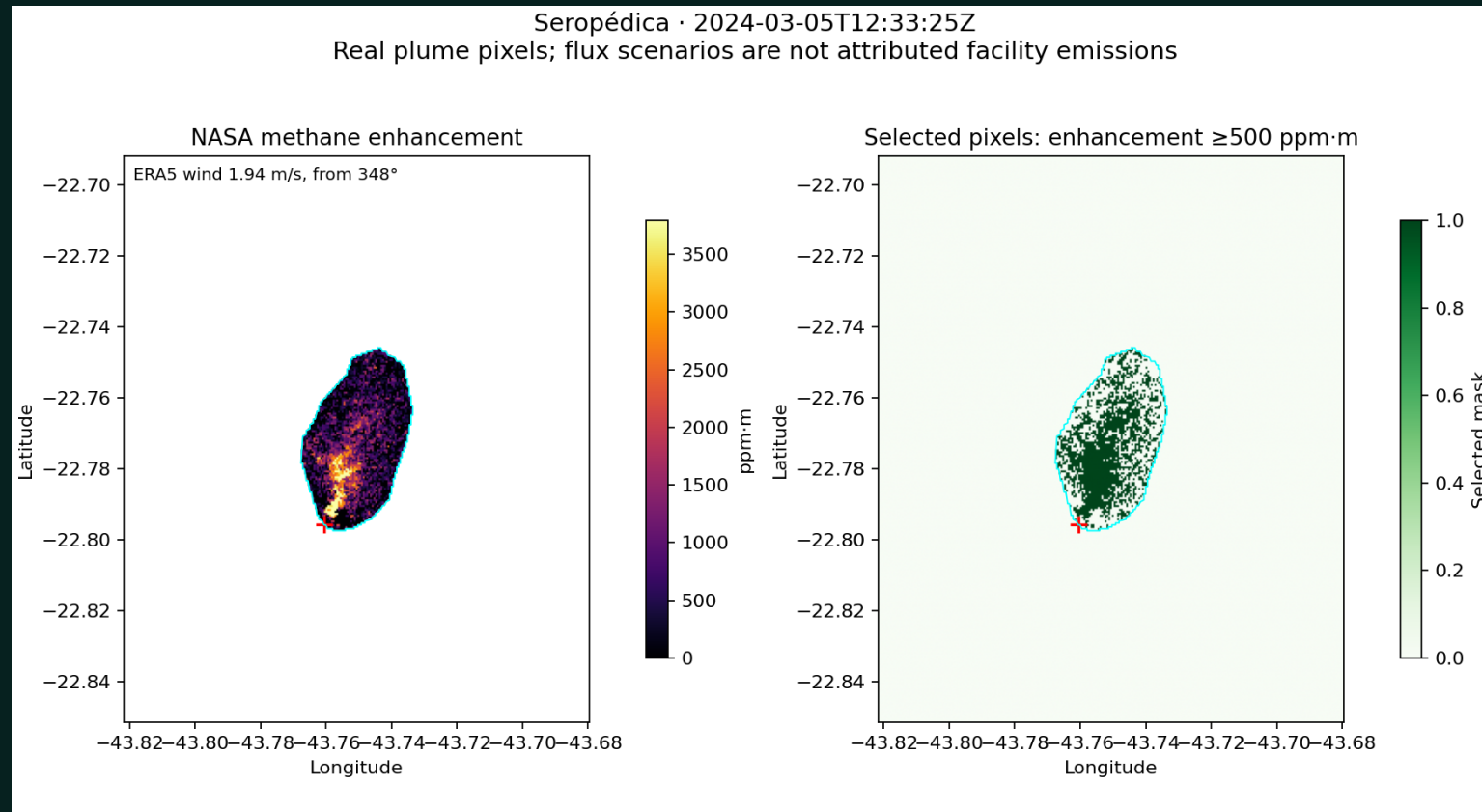
- Jeddah: 9 June 2024; Seropedica: 5 March and 24 September 2024
- Inspect pixels, overpass wind, selected excess mass and thresholds
- Diagnostic flux scenarios are not validated facility emissions

# Hyperspectral gas evidence



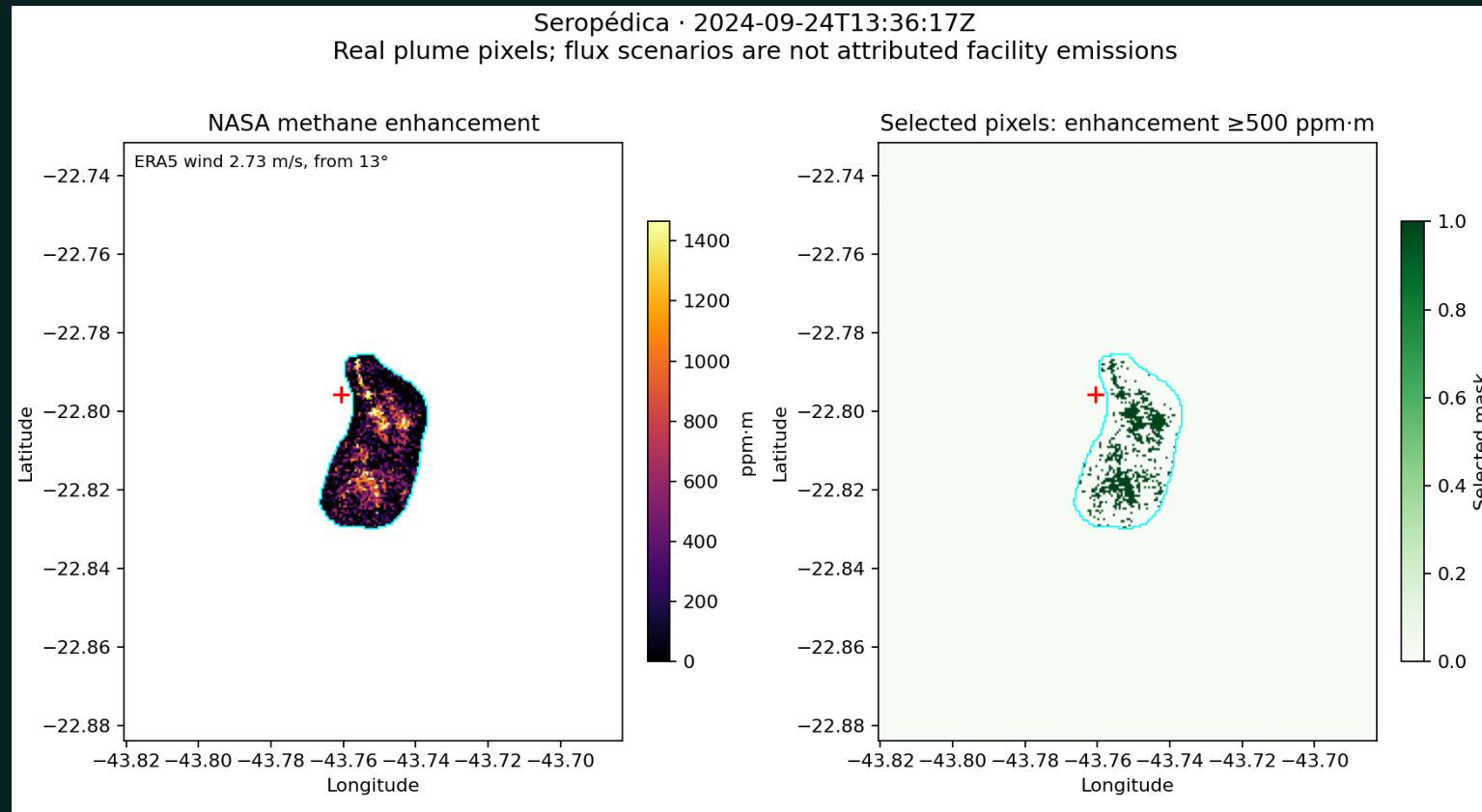
Jeddah: sensitivity correction, background subtraction and uncertainty screening.

# Brazil methane: March observation



Original NASA plume pixels; no invented sensitivity or uncertainty correction.

# Brazil methane: September observation



Real ERA5 wind and mask sensitivity; attributed emission rate remains unvalidated.

## Preprocessing: provider and project roles

- Providers supply calibrated gas retrievals and L2 surface products
- EcoOrbit applies geometry, QA, spectral bins and plume pixel selection
- EcoOrbit does not rerun radiance calibration or atmospheric correction

## Actual gas analysis

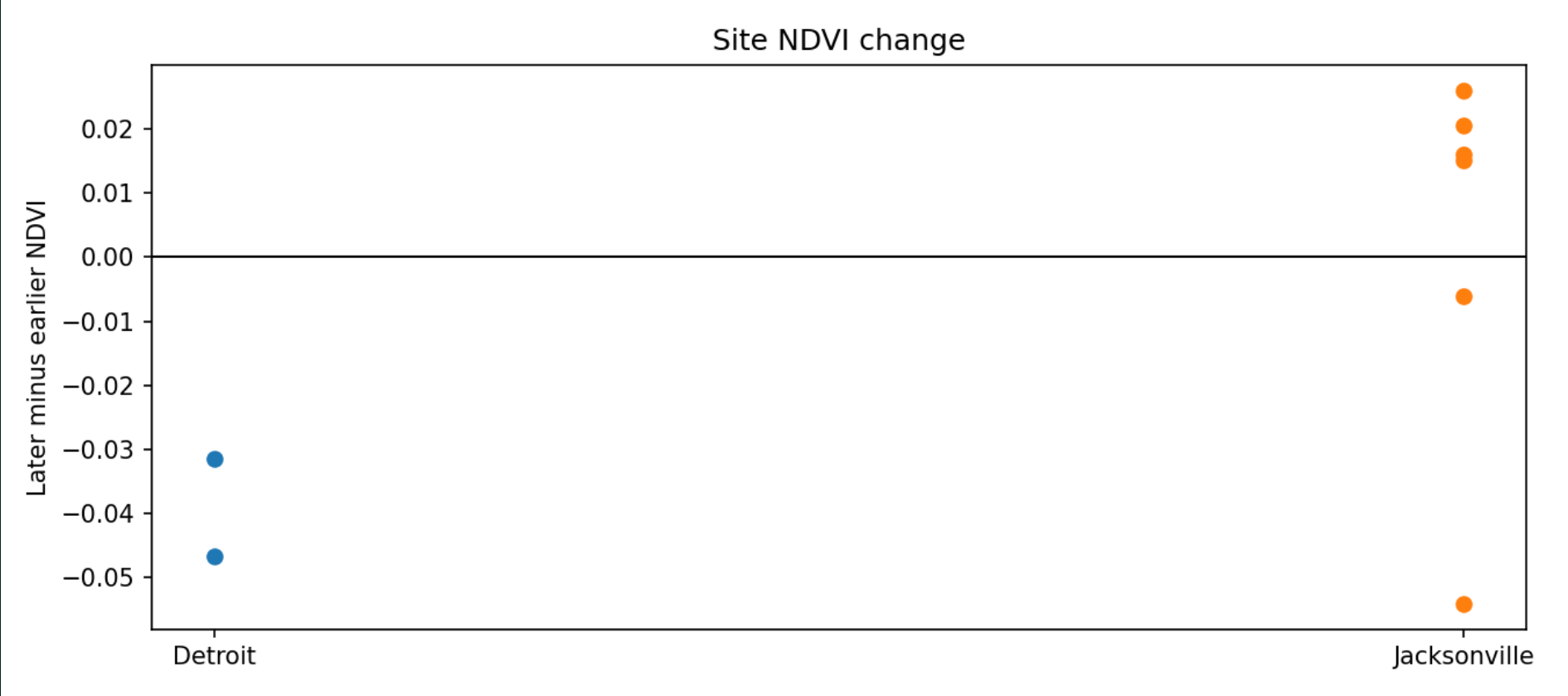
- Integrate selected column excess methane using real pixel areas
- Use pressure/temperature and overpass ERA5 wind as assumptions
- Report mask sensitivity and avoid annualizing snapshot flux

## Supplementary machine learning



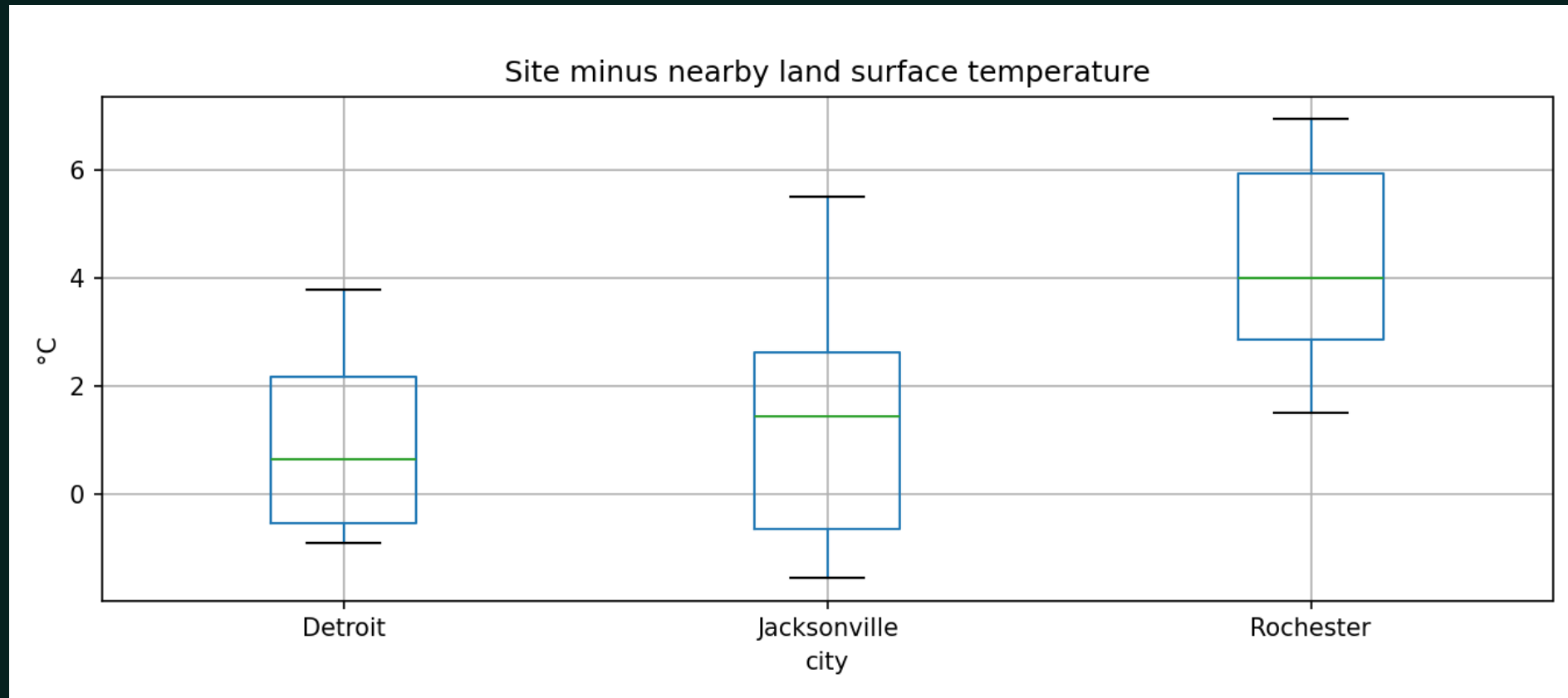
36 training / 12 test sites across all cities; 11/12 accuracy; school-control limitation.

# Multispectral site follow-up



Sentinel-2 B04/B08/B11 and SCL; completed US examples are separate from landfill events.

## Additional features



Landsat surface heat, green context, site maps and relevant NO<sub>2</sub>/SO<sub>2</sub> columns.

## Carbon evidence and reporting

- Published Tanager CO2 reference: 838,000 kg/h at power-generation site
- This is provider-reported evidence, not an EcoOrbit landfill retrieval
- GRI 305 / GHG Protocol: no annual inventory without facility records

# Judge walkthrough and reproduction

- Website pages: Solution, Methodology, Additional features
- Run `src.emit_cases`, `src.evaluate_sites` and `src.build_website`
- Real bundled inputs, notebooks and precise source/processing documentation